

JIAYING HUI

Hong Kong

MSc Artificial Intelligence & Business Analytics

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PROFILE

AI graduate student with hands-on experience in LLM-based systems, RAG architecture, and machine learning optimization. Experienced in designing scalable AI applications and leading technical projects.

EDUCATION

Lingnan University, Hong Kong

2025 – Present

MSc in Artificial Intelligence & Business Analytics

Relevant Coursework: Large Language Models, Practical Deep Learning, Machine Learning for Business, Big Data Analytics, Healthcare Analytics, Data Visualization

Wenhua College

2021 – 2025

BEng in Computer Science

Core Coursework: Data Structures, Operating Systems, Computer Networks, Database Systems, Algorithm Design

Final Year Project: Smart Logistics Management System (SpringBoot-based)

TECHNICAL SKILLS

Languages: Python, SQL

AI/ML: LLM, RAG, Multi-Agent Systems, CNN, Gradient Boosting, Random Forest

Frameworks: PyTorch, OpenCV

Databases: PostgreSQL, MongoDB

Concepts: Prompt Engineering, Task Decomposition, Model Evaluation (F1, Precision, Recall)

PROJECT EXPERIENCE

Campus Policy Intelligent QA Platform (RAG + Agent System)

- Designed full RAG pipeline: retrieval → re-ranking → generation
- Built vector-based knowledge retrieval workflow
- Developed structured prompt templates to reduce hallucination
- Implemented agent orchestration logic
- Led system architecture design and team coordination

Hong Kong Travel Planning Agent (Planner Agent Developer)

- Designed and implemented Planner Agent for multi-step trip planning
- Built task decomposition workflow (intent parsing → attraction selection → scheduling)
- Developed structured prompt strategy for reliable generation
- Improved multi-turn output stability

ICU Mortality Prediction (Machine Learning)

- Built Decision Tree, Random Forest, Gradient Boosting, and MLP models
- Performed parameter tuning and cross-validation
- Achieved best F1-score of 0.961 using Gradient Boosted Tree

Facial Age Recognition System (Deep Learning)

- Implemented RetinaFace + ResNet pipeline using PyTorch
- Achieved 95% LFW validation accuracy
- Developed browser-based real-time detection interface

Smart Logistics Management System (SpringBoot)

- Developed backend modules and RESTful services with database schema design
- Improved system stability and routing efficiency through load balancing and data analysis